

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.11/dist-packages (3.9.1)
```

Requirement already satisfied: click in
/usr/local/lib/python3.11/dist-packages (from nltk) (8.2.1)

```
Requirement already satisfied: joblib in
/usr/local/lib/python3.11/dist-packages (from nltk) (1.5.1)
```

Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk) (2024.11.6)

```
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from nltk) (4.67.1)
```

```
import nltk
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded = files.upload()
```

```
<IPython.core.display.HTML object>
```

Saving train data.txt to train data (1).txt

```
df = pd.read_csv("train_data.txt",
                 sep=":::",
                 names=["Movie_name", "Genre", "Description"],
                 engine="python"
                )
```

```
df.head(5)
df.describe()
```

```
{
  "summary": {
    "name": "df",
    "rows": 4,
    "fields": [
      {
        "column": "Movie_name",
        "properties": {
          "dtype": "string",
          "num_unique_values": 3,
          "samples": [
            "54214",
            "Nature's Fury: Storm of the Century (2006)",
            "1"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "Genre",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            27,
            "13613",
            "54214"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "Description",
        "properties": {
          "dtype": "string",
          "num_unique_values": 4,
          "samples": [
            54086,
            "12",
            "54214"
          ],
          "semantic_type": "",
          "description": ""
        }
      }
    ]
  },
  "type": "dataframe"
}
```

```

print(df.columns)
Index(['Movie_name', 'Genre', 'Description'], dtype='object')
df['Genre'].unique()
array([' drama ', ' thriller ', ' adult ', ' documentary ', ' comedy ',
       ' crime ', ' reality-tv ', ' horror ', ' sport ', ' animation ',
       ' action ', ' fantasy ', ' short ', ' sci-fi ', ' music ',
       ' adventure ', ' talk-show ', ' western ', ' family ', '
mystery ',
       ' history ', ' news ', ' biography ', ' romance ', ' game-show ',
       ' musical ', ' war '], dtype=object)

df['Genre'].value_counts()
Genre
drama      13613
documentary 13096
comedy      7447
short       5073
horror      2204
thriller    1591
action      1315
western     1032
reality-tv   884
family       784
adventure    775
music        731
romance      672
sci-fi       647
adult        590
crime        505
animation    498
sport        432
talk-show    391
fantasy      323
mystery      319
musical      277
biography    265
history      243
game-show    194
news         181
war          132
Name: count, dtype: int64

df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
Index: 54214 entries, 1 to 54214
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Movie_name      54214 non-null  object
1   Genre           54214 non-null  object
2   Description     54214 non-null  object
dtypes: object(3)
memory usage: 1.7+ MB

df=df[:14000]

df['Genre'].isnull().sum(),df['Description'].isnull().sum()

(np.int64(0), np.int64(0))

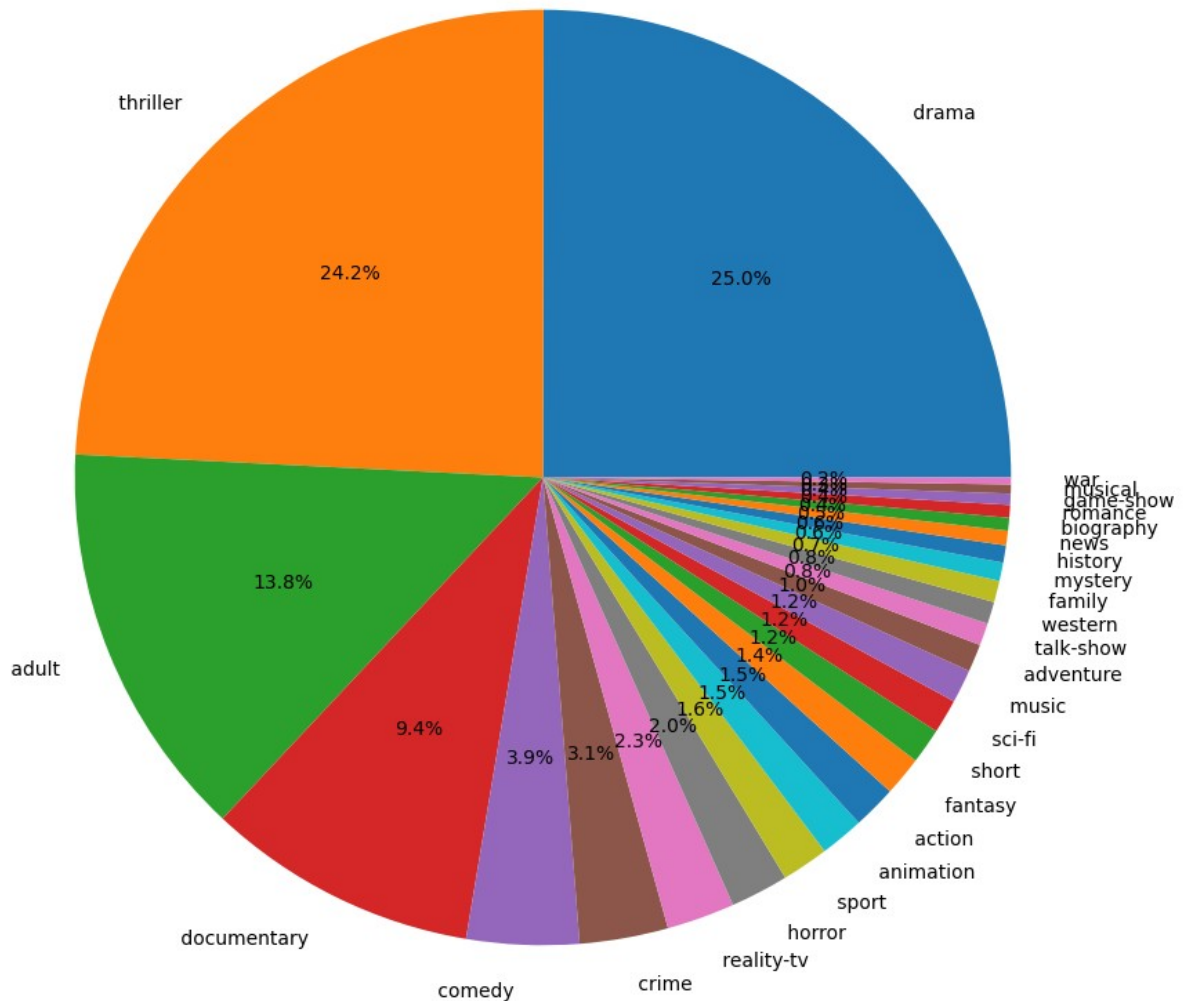
plt.figure(figsize=(11,11))
plt.pie(df['Genre'].value_counts(),labels=[' drama ', ' thriller ', '
adult ', ' documentary ', ' comedy ',
      ' crime ', ' reality-tv ', ' horror ', ' sport ', ' animation
',
      ' action ', ' fantasy ', ' short ', ' sci-fi ', ' music ',
      ' adventure ', ' talk-show ', ' western ', ' family ', '
mystery ',
      ' history ', ' news ', ' biography ', ' romance ', ' game-show
',
      ' musical ', ' war '],autopct='%0.1f%%')

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```

```
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Text(-0.7578464077547891, 0.7972884184867871, 'thriller '),
Text(-1.019912433085333, -0.41204202314564514, 'adult '),
Text(-0.4876980014718093, -0.985977007521171, 'documentary '),
Text(-0.04762522819280862, -1.0989685335074808, 'comedy '),
Text(0.19106669176357727, -1.0832790588294976, 'crime '),
Text(0.3714499095378242, -1.0353863842567868, 'reality-tv '),
Text(0.509260757544074, -0.9750146054422139, 'horror '),
Text(0.6166571819126249, -0.9108973158350947, 'sport '),
Text(0.7028767825920313, -0.8461466944289711, 'animation '),
Text(0.7797347141382024, -0.7758954669076343, 'action '),
Text(0.8464621239038461, -0.7024968845455403, 'fantasy '),
Text(0.9006737455094288, -0.6314956881483964, 'short '),
Text(0.9456835188780103, -0.561856460426063, 'sci-fi '),
Text(0.9843296826695614, -0.49101433361526275, 'music '),
Text(1.0147494817110592, -0.4245980326934367, 'adventure '),
Text(1.0362999016767536, -0.36889363478481296, 'talk-show '),
Text(1.0528847590425852, -0.3184865525824248, 'western '),
Text(1.0667788571430101, -0.26829623544256676, 'family '),
Text(1.0773947135695796, -0.22185723150784073, 'mystery '),
Text(1.0851531314354288, -0.18011852024676134, 'history '),
Text(1.0907284241938284, -0.14251843619563087, 'news '),
Text(1.0944955486850267, -0.109906750969457, 'biography '),
Text(1.0971118438432848, -0.07965928758649655, 'romance '),
Text(1.0988019858875004, -0.05132441728539519, 'game-show '),
Text(1.0996648951894925, -0.027149922428663594, 'musical '),
Text(1.0999698366746136, -0.00814606690520206, 'war ')],
[Text(0.4245495671975846, 0.42397837797739596, '25.0%'),
Text(-0.41337076786624855, 0.43488459190188383, '24.2%'),
Text(-0.5563158725919998, -0.22475019444307917, '13.8%'),
Text(-0.26601709171189597, -0.5378056404660932, '9.4%'),
Text(-0.025977397196077428, -0.5994373819131713, '3.9%'),
Text(0.10421819550740578, -0.5908794866342714, '3.1%'),
Text(0.2026090415660859, -0.5647562095946109, '2.3%'),
Text(0.27777859502404034, -0.5318261484230257, '2.0%'),
Text(0.3363584628614317, -0.49685308136459705, '1.6%'),
Text(0.38338733595928975, -0.46153456059762055, '1.5%'),
Text(0.4253098440753831, -0.42321570922234597, '1.5%'),
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Text(0.49127658845968836, -0.34445219353548895, '1.2%'),
Text(0.5158273739334601, -0.30646716023239795, '1.2%'),
Text(0.5369070996379426, -0.26782600015377966, '1.2%'),
Text(0.5534997172969414, -0.23159892692369272, '1.0%'),
```

```
Text(0.5652544918236837, -0.20121470988262521, '0.8%'),
Text(0.5743007776595918, -0.17371993777223171, '0.8%'),
Text(0.58187937662346, -0.14634340115049094, '0.7%'),
Text(0.5876698437652252, -0.12101303536791311, '0.6%'),
Text(0.5919017080556884, -0.09824646558914253, '0.6%'),
Text(0.5949427768329972, -0.07773732883398048, '0.5%'),
Text(0.5969975720100146, -0.05994913689243108, '0.4%'),
Text(0.5984246420963371, -0.04345052050172539, '0.4%'),
Text(0.5993465377568183, -0.027995136701124645, '0.4%'),
Text(0.599817215557905, -0.014809048597452868, '0.3%'),
Text(0.5999835472770619, -0.004443309221019305, '0.2%')] ]
```



```
df.tail()
```

```

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"Movie_name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \" Nieuw tiengemetten (2010) \",\n          \" Hanna D. - La ragazza del Vondel Park (1984) \",\n          \" That's What She Told Me (2011) \"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Genre\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 4,\n        \"samples\": [\n          \" documentary \",\n          \" drama \",\n          \" comedy \"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"Description\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \" New Tiengemetten is the account of the transformation of the little island of Tiengemetten, the last \\\"real\\\" island of South-West Holland. Fertile agricultural land is set aside to allow nature to run its course, according to a carefully drawn up plan. The film follows the development of the island after the departure of the last farmer. The asphalt roads are dug up to creeks, the dike is cut, the farms are given a new destination. Will man succeed in making nature? Nature that is also fit for recreation?\",\n          \" Hanna is a girl completely abandoned to herself, is not at all inhibited sexually, and is attracted to the attention of unscrupulous people like Miguel, who takes advantage of Hanna, turning her to prostitution to obtain her drug fix. Until she meets Alex, a young man with a sincere affection for Hanna.\",\n          \" A New York Artist accidentally summons the Ghost of Johnny Cash, who helps her find the father she has never met... Carly North is a NYC painter, whose love affair with a wanna-be Naked Singing Cowboy and creative life are at a complete standstill. After a visit from the Ghost of Johnny Cash, she is propelled on a surreal trip throughout the wilds of Upstate New York to find the father she has never met; and some answers to a few life altering questions. THAT'S WHAT SHE TOLD ME explores the ideas of fantasy infringing on reality, and the struggle to decipher what is real and what isn't. Using music, an American icon, and the lure of the road; this story speaks to anyone who has ever wondered where they really come from and whether we like it or not: the indelible mark of family on our lives.\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ],\n  \"type\": \"dataframe\"}

def remove_punct(text):
    import string as st
    punct=st.punctuation
    new_text=''.join([char for char in text if char not in punct])
    return new_text

df['Description']=df['Description'].apply(lambda x:remove_punct(x))

df.tail()

```

```

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"Movie_name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 5,\n        \"samples\": [\n          \" Nieuw tiengemetten (2010) \",\n          \" Hanna D. - La ragazza del Vondel Park (1984) \",\n          \" That's What She Told Me (2011) \",\n          \" New Tiengemetten is the account of the transformation of the little island of Tiengemetten the last real island of SouthWest Holland Fertile agricultural land is set aside to allow nature to run its course according to a carefully drawn up plan The film follows the development of the island after the departure of the last farmer The asphalt roads are dug up to creeks the dike is cut the farms are given a new destination Will man succeed in making nature Nature that is also fit for recreation\",\n          \" Hanna is a girl completely abandoned to herself is not at all inhibited sexually and is attracted to the attention of unscrupulous people like Miguel who takes advantage of Hanna turning her to prostitution to obtain her drug fix Until she meets Alex a young man with a sincere affection for Hanna\",\n          \" A New York Artist accidentally summons the Ghost of Johnny Cash who helps her find the father she has never met Carly North is a NYC painter whose love affair with a wannabe Naked Singing Cowboy and creative life are at a complete standstill After a visit from the Ghost of Johnny Cash she is propelled on a surreal trip throughout the wilds of Upstate New York to find the father she has never met and some answers to a few life altering questions THATS WHAT SHE TOLD ME explores the ideas of fantasy infringing on reality and the struggle to decipher what is real and what isnt Using music an American icon and the lure of the road this story speaks to anyone who has ever wondered where they really come from and whether we like it or not the indelible mark of family on 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is attracted to the attention of unscrupulous people like Miguel who takes advantage of Hanna turning her to prostitution to obtain her drug fix Until she meets Alex a young man with a sincere affection for Hanna\",\n          \" A New York Artist accidentally summons the Ghost of Johnny Cash who helps her find the father she has never met Carly North is a NYC painter whose love affair with a wannabe Naked Singing Cowboy and creative life are at a complete standstill After a visit from the Ghost of Johnny Cash she is propelled on a surreal trip throughout the wilds of Upstate New York to find the father she has never met and some answers to a few life altering questions THATS WHAT SHE TOLD ME explores the ideas of fantasy infringing on reality and the struggle to decipher what is real and what isnt Using music an American icon and the lure of the road this story speaks to anyone who has ever wondered where they really come from and whether we like it or not the indelible mark of family on our lives\",\n          \" New York Artist accidentally summons the Ghost of Johnny Cash who helps her find the father she has never met Carly North is a NYC painter whose love affair with a wannabe Naked Singing Cowboy and creative life are at a complete standstill After a visit from the Ghost of Johnny Cash she is propelled on a surreal trip throughout the wilds of Upstate New York to find the father she has never met and some answers to a few life altering questions THATS WHAT SHE TOLD ME explores the ideas of fantasy infringing on reality and the struggle to decipher what is real and what isnt Using music an American icon and the lure of the road this story speaks to anyone who has ever wondered where they really come from and whether we like it or not the indelible mark of family on our lives\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}","type":"dataframe"}

```

```

df['Description']=df['Description'].apply(lambda x:x.lower())
df.head()

```

```

{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 14000,\n  \"fields\": [\n    {\n      \"column\": \"Movie_name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 14000,\n        \"samples\": [\n          \" Chepachet (2007) \",\n          \" Fleeting (2017/II) \",\n          \" Report to New York (1959) \",\n          \" New York Artist accidentally summons the Ghost of Johnny Cash who helps her find the father she has never met Carly North is a NYC painter whose love affair with a wannabe Naked Singing Cowboy and creative life are at a complete standstill After a visit from the Ghost of Johnny Cash she is propelled on a surreal trip throughout the wilds of Upstate New York to find the father she has never met and some answers to a few life altering questions THATS WHAT SHE TOLD ME explores the ideas of fantasy infringing on reality and the struggle to decipher what is real and what isnt Using music an American icon and the lure of the road this story speaks to anyone who has ever wondered where they really come from and whether we like it or not the indelible mark of family on our lives\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    }\n  ]\n}","type":"dataframe"}

```



```

"description\": \"\"\\n      }\\n    },\\n    {\\n      \"column\\\":
\"Genre\\\",\\n      \"properties\\\": {\\n        \"dtype\\\": \"category\\\",\\n
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        \" sport \\\",\\n        \" sci-fi \\\",\\n        \" animation \\\"\\n
        ],\\n        \"semantic_type\\\": \"\\\",\\n        \"description\\\": \"\"\\n
        }\\n      },\\n      {\\n        \"column\\\": \"Description\\\",\\n
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        \"num_unique_values\\\": 13987,\\n          \"samples\\\": [\\n            \" a
race against time is ongoing in malaysia conservationists are on a
desperate mission to save the countrys last sumatran rhinos the only
way to achieve this is to capture all remaining individuals in the
wild a dangerous and costly operation operation sumatran rhino
chronicles malaysias foremost rhino scientists in their incredible
quest to save this critically endangered species\\\",\\n            \"
based on leslea newmans story a letter to harvey milk this
awardwinning short examines the sensitive issues that surface when
harry weinberg an older jewish man enrolls in a creative writing class
taught by a lesbian teacher for one particular assignment harry writes
a letter to his former customer and friend the slain political
activist harvey milk the letter unleashes a series of events which
impel marvin harrys best friend to painfully recall and reveal a
tragic incident that occurred in a concentration camp forty years
earlier\\\",\\n            \" the series is mainly about a 27 year old man
named makio sakaki his purpose is to become a successor after his
father however he must pass high school first in order to prove
himself worthy in the family for if he fails his younger brother will
take over instead of him through the school years he faces many
obstacles such as righteousness friendship and perhaps even love\"\\n
        ],\\n        \"semantic_type\\\": \"\\\",\\n        \"description\\\": \"\"\\n
        }\\n      }\\n    ]\\n  }\", \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```

from nltk.tokenize import TreebankWordTokenizer

```

```

tokenizer = TreebankWordTokenizer()

```

```

def tokenize(text):
    if isinstance(text, list): # if already tokenized
        return text
    elif isinstance(text, str): # if it's a string, tokenize it
        return tokenizer.tokenize(text)
    else:
        return [] # fallback for None or other types

```

```

df['Description'] = df['Description'].fillna('').apply(toknize)

```

```

df.head(5)

```

```

{"summary": "{\\n  \"name\\\": \"df\\\",\\n  \"rows\\\": 14000,\\n  \"fields\\\":
[\\n    {\\n      \"column\\\": \"Movie_name\\\",\\n      \"properties\\\": {\\n
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```



```

\"samples\": [\n          \" Chepachet (2007) \",\n          \"Fleeting (2017/II) \",\n          \" \\\\\"Report to New York\\\\\" (1959)\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\":\n        \"Genre\",\n        \"properties\": {\n          \"dtype\": \"category\",\n          \"num_unique_values\": 27,\n          \"samples\": [\n            \" sport \",\n            \" sci-fi \",\n            \" animation \"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"Description\",\n          \"properties\": {\n            \"dtype\": \"object\",\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          }\n        }\n      ]\n    },\n    \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```

def stemming(text):
    from nltk import PorterStemmer
    ps=PorterStemmer().stem
    # dir(PorterStemmer())
    new_text=" ".join([ps(word) for word in text ])
    return new_text
df['Description']=df['Description'].apply(lambda x : stemming(x))

```

```
df.head(5)
```

```

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 14000,\n  \"fields\":\n  [\n    {\n      \"column\": \"Movie_name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 14000,\n        \"samples\": [\n          \" Chepachet (2007) \",\n          \"Fleeting (2017/II) \",\n          \" \\\\\"Report to New York\\\\\" (1959)\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\":\n        \"Genre\",\n        \"properties\": {\n          \"dtype\": \"category\",\n          \"num_unique_values\": 27,\n          \"samples\": [\n            \" sport \",\n            \" sci-fi \",\n            \" animation \"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"Description\",\n          \"properties\": {\n            \"dtype\": \"string\",\n            \"num_unique_values\": 13987,\n            \"samples\": [\n              \"a\nrace against time is ongo in malaysia conservationist are on a desper\nmission to save the countri last sumatran rhino the onli way to achiev\nthi is to captur all remain individu in the wild a danger and costli\noper oper sumatran rhino chronicl malaysia foremost rhino scientist in\ntheir incred quest to save thi critic endang speci\",\n              \"base on leslea newman stori a letter to harvey milk thi awardwin\nshort examin the sensit issu that surfac when harri weinberg an older\njewish man enrol in a creativ write class taught by a lesbian teacher\nfor one particular assign harri write a letter to hi former custom and\nfriend the slain polit activist harvey milk the letter unleash a seri\nof event which impel marvin harri best friend to pain recal and reveal\na tragic incid that occur in a concentr camp forti year earlier\",\n              \"the seri is mainli about a 27 year old man name makio sakaki hi

```

```

purpos is to becom a successor after hi father howev he must pass high
school first in order to prove himself worthi in the famili for if he
fail hi younger brother will take over instead of him through the
school year he face mani obstacl such as righteous friendship and
perhap even love\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n          }\\n          ]\\
n}\\\", \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```

from sklearn.feature_extraction.text import TfidfVectorizer

```

```

tf_idf=TfidfVectorizer()

```

```

Description_tf_idf=tf_idf.fit_transform(df['Description'])

```

```

print(Description_tf_idf.shape)

```

```

Description_tf_idf=pd.DataFrame(Description_tf_idf.toarray())

```

```

Description_tf_idf.columns = tf_idf.get_feature_names_out()

```

```

Description_tf_idf

```

```

(14000, 50500)

```

```

{\"type\": \"dataframe\", \"variable_name\": \"Description_tf_idf\"}

```

```

df['Description len']=df['Description'].apply(lambda x:len(x)-
x.count(' '))
df.head(10)

```

```

{\"summary\": \"{\\n  \\\"name\\\": \\\"df\\\",\\n  \\\"rows\\\": 14000,\\n  \\\"fields\\\":
[\\n    {\\n      \\\"column\\\": \\\"Movie_name\\\",\\n      \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"string\\\",\\n      \\\"num_unique_values\\\": 14000,\\n
\\\"samples\\\": [\\n        \\\" Chepachet (2007) \\\",\\n        \\\"
Fleeting (2017/II) \\\",\\n        \\\" \\\"\\\"Report to New York\\\"\\\" (1959)
\\\"\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n    },\\n    {\\n      \\\"column\\\":
\\\"Genre\\\",\\n      \\\"properties\\\": {\\n        \\\"dtype\\\": \\\"category\\\",\\n
        \\\"num_unique_values\\\": 27,\\n        \\\"samples\\\": [\\n
\\\" sport \\\",\\n        \\\" sci-fi \\\",\\n        \\\" animation \\\"\\n
      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
    }\\n  },\\n  {\\n    \\\"column\\\": \\\"Description\\\",\\n
\\\"properties\\\": {\\n      \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 13987,\\n      \\\"samples\\\": [\\n        \\\"a
race against time is ongo in malaysia conservationist are on a desper
mission to save the countri last sumatran rhino the onli way to achiev
thi is to captur all remain individu in the wild a danger and costli
oper oper sumatran rhino chronicl malaysia foremost rhino scientist in
their incred quest to save thi critic endang speci\\\",\\n
\\\"base on leslea newman stori a letter to harvey milk thi awardwin
short examin the sensit issu that surfac when harri weinberg an older
jewish man enrol in a creativ write class taught by a lesbian teacher

```

```

for one particular assign harri write a letter to hi former custom and
friend the slain polit activist harvey milk the letter unleash a seri
of event which impel marvin harri best friend to pain recal and reveal
a tragic incid that occur in a concentr camp forti year earlier\",\\n
\\nthe seri is mainli about a 27 year old man name makio sakaki hi
purpos is to becom a successor after hi father howev he must pass high
school first in order to prove himself worthi in the famili for if he
fail hi younger brother will take over instead of him through the
school year he face mani obstacl such as righteous friendship and
perhap even love\\n\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n          \\\"column\\\":
\\\"Description len\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\":
\\\"number\\\",\\n          \\\"std\\\": 319,\\n          \\\"min\\\": 33,\\n
\\\"max\\\": 7483,\\n          \\\"num_unique_values\\\": 1422,\\n
\\\"samples\\\": [\\n          824,\\n          744,\\n          398\\n
],\\n          \\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n
}\\n          }\\n          ]\\n}\\",\"type\":\"dataframe\",\"variable_name\":\"df\"}

```

```

from sklearn.preprocessing import MinMaxScaler
scaler=MinMaxScaler()
df['Description_len_sclr'] = scaler.fit_transform(df['Description
len'].values.reshape(-1, 1))
df.head(5)

```

```

{"summary":{"\\n \\\"name\\\": \\\"df\\\",\\n \\\"rows\\\": 14000,\\n \\\"fields\\\":
[\\n      {\\n          \\\"column\\\": \\\"Movie_name\\\",\\n          \\\"properties\\\": {\\n
\\\"dtype\\\": \\\"string\\\",\\n          \\\"num_unique_values\\\": 14000,\\n
\\\"samples\\\": [\\n          \\\" Chepachet (2007) \\\",\\n          \\\"
Fleeting (2017/II) \\\",\\n          \\\" \\\"\\\"Report to New York\\\"\\\" (1959)
\\\"\\n          ],\\n          \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n          }\\n          },\\n          {\\n          \\\"column\\\":
\\\"Genre\\\",\\n          \\\"properties\\\": {\\n          \\\"dtype\\\": \\\"category\\\",\\n
          \\\"num_unique_values\\\": 27,\\n          \\\"samples\\\": [\\n
\\\" sport \\\",\\n          \\\" sci-fi \\\",\\n          \\\" animation \\\"\\n
],\\n          \\\"semantic_type\\\": \\\"\\\",\\n          \\\"description\\\": \\\"\\\"\\n
}\\n          },\\n          {\\n          \\\"column\\\": \\\"Description\\\",\\n
\\\"properties\\\": {\\n          \\\"dtype\\\": \\\"string\\\",\\n
\\\"num_unique_values\\\": 13987,\\n          \\\"samples\\\": [\\n          \\\"a
race against time is ongo in malaysia conservationist are on a desper
mission to save the countri last sumatran rhino the onli way to achiev
thi is to captur all remain individu in the wild a danger and costli
oper oper sumatran rhino chronicl malaysia foremost rhino scientist in
their incred quest to save thi critic endang speci\\\",\\n
\\\"base on leslea newman stori a letter to harvey milk thi awardwin
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for one particular assign harri write a letter to hi former custom and
friend the slain polit activist harvey milk the letter unleash a seri
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a tragic incid that occur in a concentr camp forti year earlier\\\",\\n

```

```

\"the seri is mainli about a 27 year old man name makio sakaki hi
purpos is to becom a successor after hi father howev he must pass high
school first in order to prove himself worthi in the famili for if he
fail hi younger brother will take over instead of him through the
school year he face mani obstacl such as righteous friendship and
perhap even love\"\\n
    ],\\n
    \"semantic_type\\\": \"\\\",\\n
\"description\\\": \"\\\"\\n
    }\\n
    },\\n
    {\\n
    \"column\\\":
\"Description len\\\",\\n
    \"properties\\\": {\\n
    \"dtype\\\":
\"number\\\",\\n
    \"std\\\": 319,\\n
    \"min\\\": 33,\\n
\"max\\\": 7483,\\n
    \"num_unique_values\\\": 1422,\\n
\"samples\\\": [\\n
    824,\\n
    744,\\n
    398\\n
    ],\\n
    \"semantic_type\\\": \"\\\",\\n
    \"description\\\": \"\\\"\\n
    }\\n
    },\\n
    {\\n
    \"column\\\": \"Description_len_sclr\\\",\\n
\"properties\\\": {\\n
    \"dtype\\\": \"number\\\",\\n
    \"std\\\":
0.04289764873673808,\\n
    \"min\\\": 0.0,\\n
    \"max\\\":
0.9999999999999999,\\n
    \"num_unique_values\\\": 1422,\\n
\"samples\\\": [\\n
    0.10617449664429529,\\n
    0.09543624161073826,\\n
    0.04899328859060402\\n
    ],\\n
    \"semantic_type\\\": \"\\\",\\n
    \"description\\\": \"\\\"\\n
    }\\n
    }\\n
    ]\\n}\"},\"type\":\"dataframe\",\"variable_name\":\"df\"}

```

```

target=df['Genre']
feature=df.drop(['Genre','Movie_name','Description','Description
len'],axis=1)
target.head(5)

```

```

1      drama
2      thriller
3      adult
4      drama
5      drama
Name: Genre, dtype: object

```

target

```

1      drama
2      thriller
3      adult
4      drama
5      drama
...
13996      comedy
13997      documentary
13998      short
13999      documentary
14000      drama
Name: Genre, Length: 14000, dtype: object

```

feature

```
{
  "summary": "{
    \"name\": \"feature\",
    \"rows\": 14000,
    \"fields\": [
      {
        \"column\": \"Description_len_sclr\",
        \"properties\": {
          \"dtype\": \"number\",
          \"std\": 0.04289764873673808,
          \"min\": 0.0,
          \"max\": 0.9999999999999999,
          \"num_unique_values\": 1422,
          \"samples\": [
            0.10617449664429529,
            0.09543624161073826,
            0.04899328859060402
          ],
          \"semantic_type\": \"\",
          \"description\": \"\"
        }
      ]
    ],
    \"type\": \"dataframe\",
    \"variable_name\": \"feature\"
  }
```

```
target.shape , feature.shape
```

```
((14000,), (14000, 1))
```

```
Description_tf_idf.head()
```

```
{
  \"type\": \"dataframe\",
  \"variable_name\": \"Description_tf_idf\"
}
```

```
print(feature.shape)
```

```
print>Description_tf_idf.shape)
```

```
(14000, 1)
```

```
(14000, 50500)
```

```
feature.reset_index(drop=True, inplace=True)
```

```
Description_tf_idf.reset_index(drop=True, inplace=True)
```

```
feature = pd.concat([feature, pd.DataFrame>Description_tf_idf)],
axis=1)
```

```
feature.shape
```

```
(14000, 50501)
```

```
feature.head()
```

```
{
  \"type\": \"dataframe\",
  \"variable_name\": \"feature\"
}
```

```
# Suppose your DataFrame is named df
```

```
# And you want to use 'Description' as input, and 'Genre' as target
```

```
import pandas as pd
```

```
# Load the dataset again – adjust path if needed
```

```
df = pd.read_csv("train_data.txt", sep=":::", names=["Movie_name",
"Genre", "Description"], engine="python")
```

```
X = df['Description']      # Features (tokenized or raw)
```

```
y = df['Genre']           # Labels / target variable
```

```
from sklearn.model_selection import train_test_split
```

```
# Now split the data
```

```

x_train, x_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

print('x_train:',x_train.shape,'x_test:',x_test.shape,'y_train:',y_train.shape,'y_test:',y_test.shape)

x_train: (43371,) x_test: (10843,) y_train: (43371,) y_test: (10843,)

from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix
from sklearn.model_selection import KFold, cross_val_score

model=RandomForestClassifier(n_estimators=150,max_depth=50)

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.pipeline import Pipeline
model = Pipeline([
    ('tfidf', TfidfVectorizer(stop_words='english')),
    ('clf', MultinomialNB())
])

# 5. Train the model
model.fit(x_train, y_train)

# 6. Evaluate
accuracy = model.score(x_test, y_test)
print("Model Accuracy:", accuracy)

Model Accuracy: 0.4464631559531495

model.score(x_train,y_train)

0.4720896451545964

from joblib import dump

dump(model, 'Random_forest_model.joblib')

['Random_forest_model.joblib']

```